NYU TANDON SCHOOL OF ENGINEERING

AI4CE

Robot Perception

Multi-View Geometry

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ROB-GY 6203, Fall 2024

Overview

* Hands-on: AprilTag & camera calibration

+ Epipolar geometry

++ Fundamental matrix

+ Essential matrix

+ Planar Homography

+ PnP problem

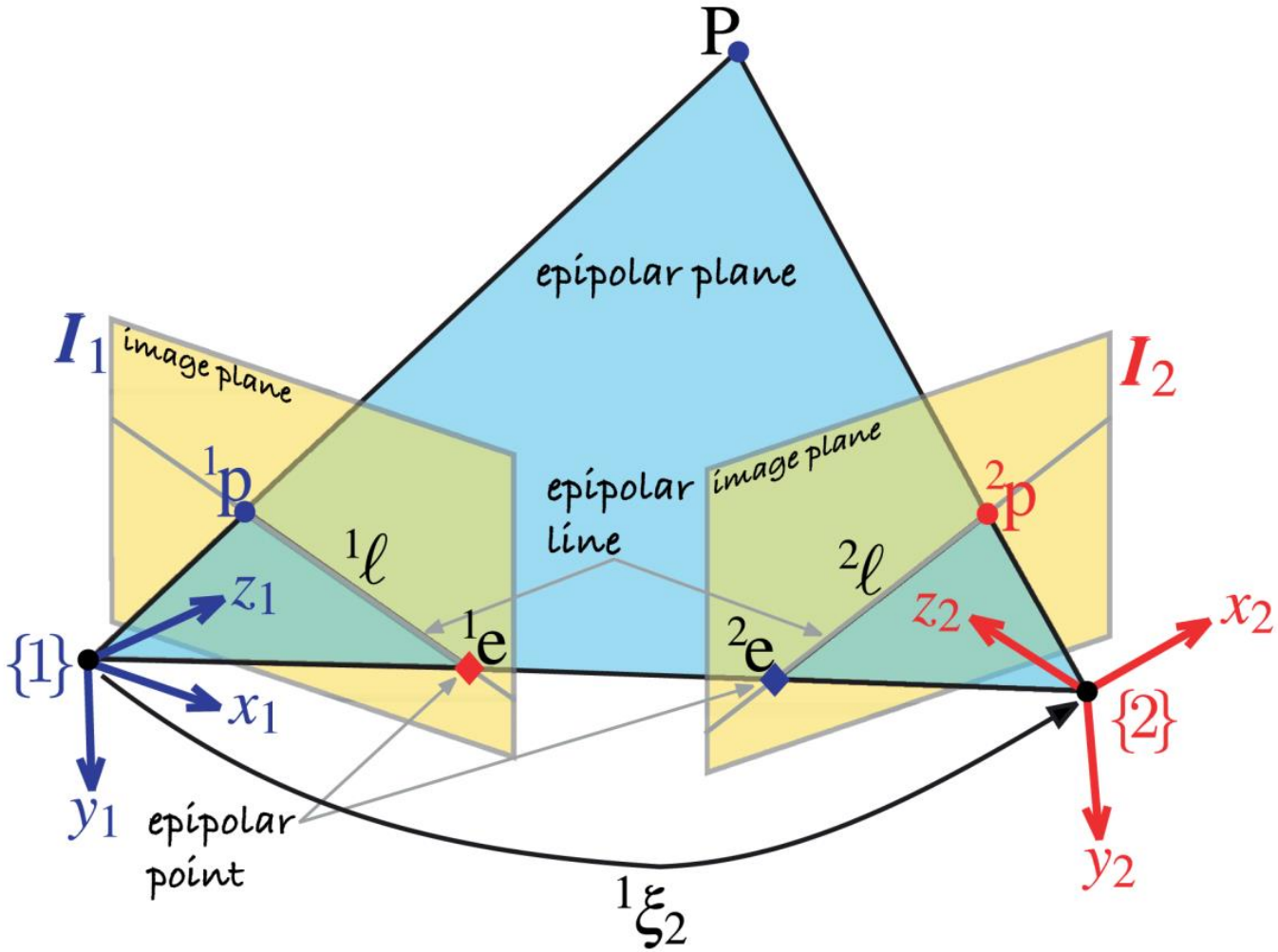
++ Hand-eye calibration

*: know how to code
++: know how to derive
+: know the concept

References

- HZ2003:
  - Section 9.1, 9.2, 9.3, 9.5, 9.6, 11.1, 11.2, 11.7
- Co2017:
  - Section 14.2, 11.2.3
- Sz2022:
  - Section 11.3, 11.2, 12.1
- FP2011:
  - Section 7.1, 8.1.2
- Radu Horaud, Fadi Dornaika. Hand-eye Calibration. International Journal of Robotics Research, SAGE Publications, 1995, 14 (3), pp.195–210.

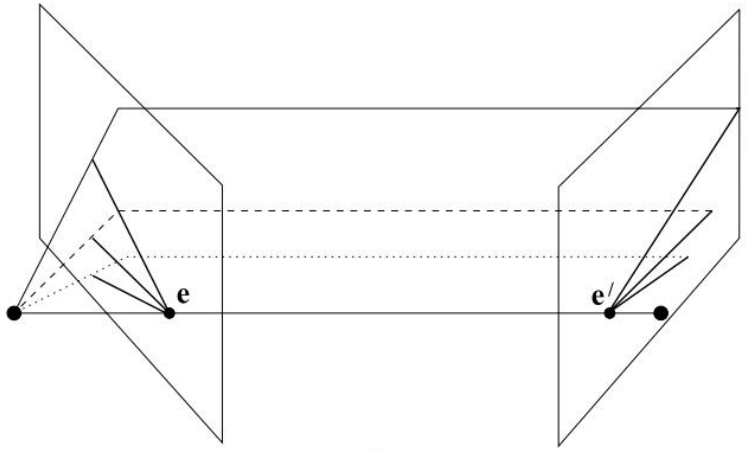
Two-view Geometry



Corke 2011

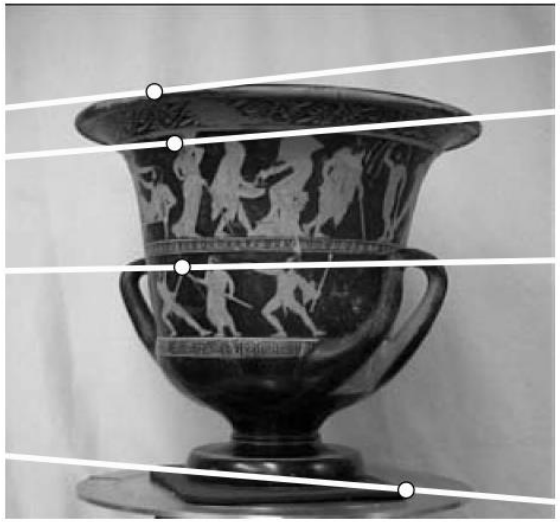
Epipolar Geometry

- **Independent of scene structure**
- Only depends on the **cameras' internal parameters and relative pose**



a

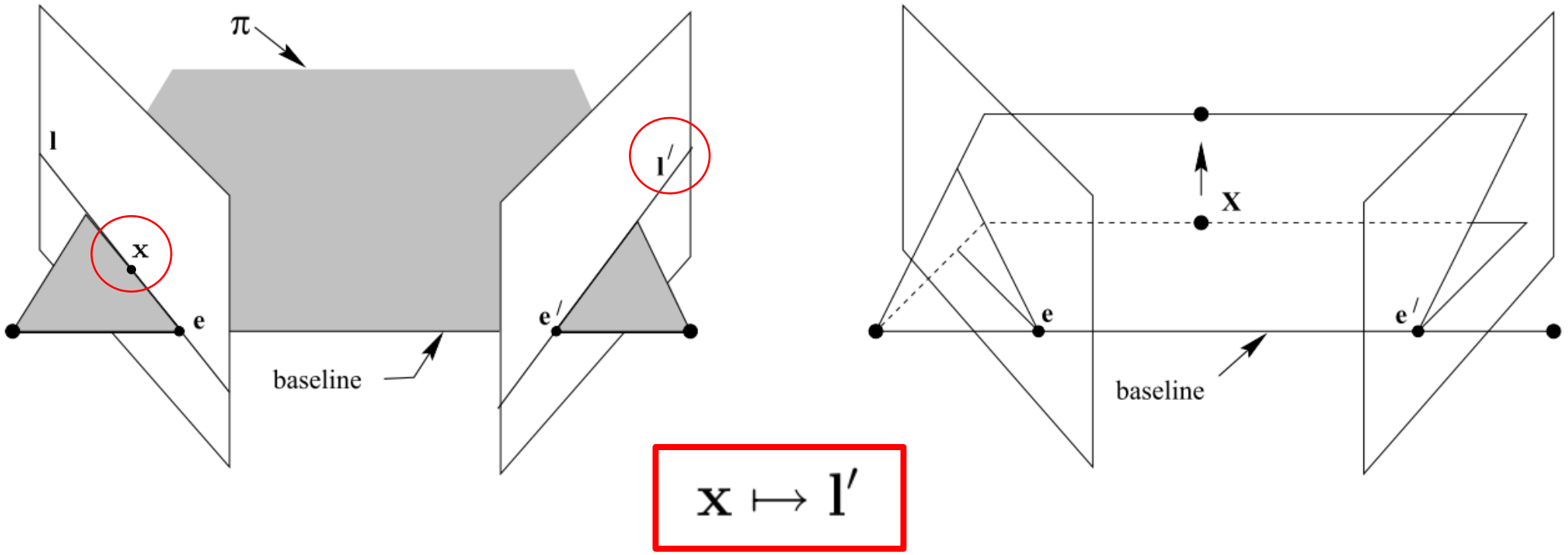




Hartley & Zisserman 2003

Epipolar Geometry

- From **a point x** on one image, epipolar geometry allows us to find on the other image **a corresponding line l'** that **MUST contain** the **corresponding point x', without knowing where x' is**



$$x \mapsto l'$$

Hartley & Zisserman 2003

Fundamental Matrix: the Algebra of Epipolar Geometry

Result 9.3. *The fundamental matrix satisfies the condition that for any pair of corresponding points* $\mathbf{x} \leftrightarrow \mathbf{x}'$ *in the two images*

$$\mathbf{x}'^{\mathsf{T}}\mathbf{F}\mathbf{x} = 0.$$

[LonguetHiggins-81] H. C. Longuet-Higgins. A computer algorithm for reconstructing a scene from two projections. *Nature*, 293:133–135, September 1981.

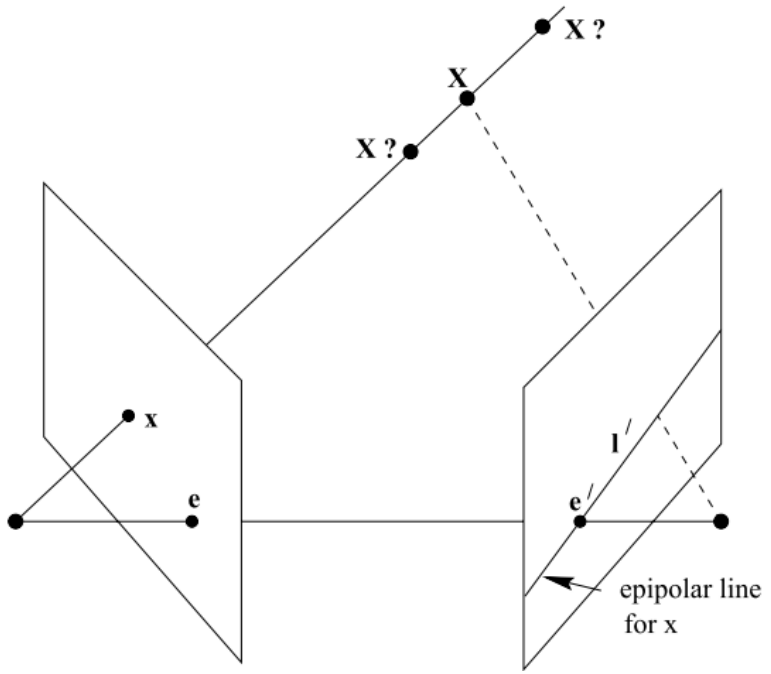
Hartley & Zisserman 2003

Fundamental Matrix: the Algebra of Epipolar Geometry



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$$\mathbf{x} \mapsto \mathbf{l}'$$

Hartley & Zisserman 2003

Important Properties of Fundamental Matrix

- $\boldsymbol{F}$ matrix is homogeneous
- $rank(\boldsymbol{F}) == 2$ ⟺

> Geometrically, F represents a mapping from the 2-dimensional projective plane $\mathbb{P}^2$ of the first image to the pencil of epipolar lines through the epipole e'. Thus, it represents a mapping from a 2-dimensional onto a 1-dimensional projective space, and hence must have rank 2.

- $\boldsymbol{F}$ matrix has **only 7** degrees-of-freedom (DOF)
  - 8 ratios – 1 rank deficiency

- $\boldsymbol{x} \Longrightarrow \boldsymbol{l'} = \boldsymbol{F}\boldsymbol{x}$
- $\boldsymbol{x'} \Longrightarrow \boldsymbol{l} = \boldsymbol{F}^T\boldsymbol{x'}$ ⟺

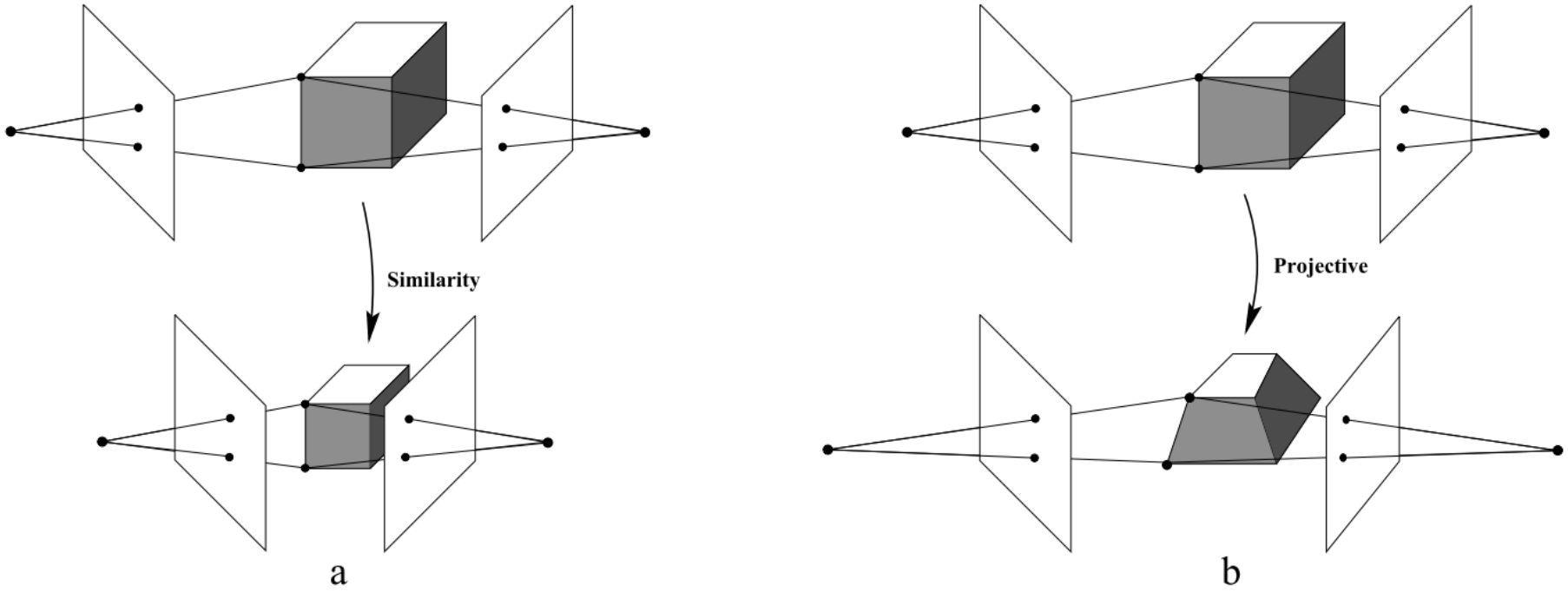
  - From $\boldsymbol{x}$ in image 1, we can use $\boldsymbol{F}$ to compute the corresponding epipolar line $\boldsymbol{l'}$ in image 2.
  - And vice versa.

- $\boldsymbol{F}\boldsymbol{e} = \boldsymbol{0}$
- $\boldsymbol{F}^T\boldsymbol{e'} = \boldsymbol{0}$ ⟺

  - Given $\boldsymbol{F}$, we can compute the epipole $\boldsymbol{e}$ in image 1 as the null space of $\boldsymbol{F}$.
  - And the epipole $\boldsymbol{e'}$ in image 2 is the left null space of $\boldsymbol{F}$.

Hartley & Zisserman 2003

Projective Ambiguity of Fundamental Matrix



Result 9.8. *If* $\mathtt{H}$ *is a* 4×4 *matrix representing a projective transformation of 3-space, then the fundamental matrices corresponding to the pairs of camera matrices* $(\mathtt{P}, \mathtt{P}')$ *and* $(\mathtt{PH}, \mathtt{P}'\mathtt{H})$ *are the same.*

How can we reduce such ambiguities?

Hartley & Zisserman 2003

Normalized Coordinate and Essential Matrix

$$\mathbf{x} = \mathrm{P}\mathbf{X}$$

$$\mathrm{P} = \mathrm{K}[\mathrm{R} \mid \mathbf{t}]$$

Normalized coordinate $$\hat{\mathbf{x}} = \mathrm{K}^{-1}\mathbf{x} = [\mathrm{R} \mid \mathbf{t}]\mathbf{X}$$

$$\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$$

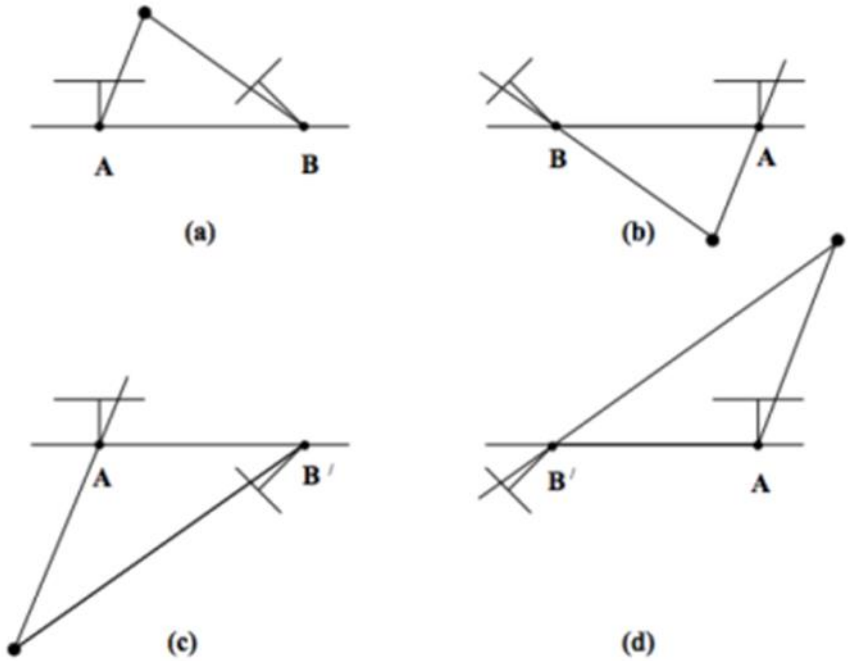
Result 9.17. *A 3×3 matrix is an essential matrix if and only if two of its singular values are equal, and the third is zero.*

$$\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K} \qquad \boxed{\mathrm{E} = [\mathbf{t}]_{\times}\mathrm{R}}$$

Hartley & Zisserman 2003

Extracting Relative Camera Pose from E-matrix

- There are 4 potential (R, t) solutions that would give the same E-matrix
  - Two potential solutions for R
  - Two potential solutions for t
- But only one solution is practical!
- How do we eliminate the impossible?
  - Using our prior knowledge
- All 3D points should be in front of the two cameras
  - Reconstruction 3D points for each potential solution
  - Count #points (=N) in front of the camera
  - Choose the potential solution with the max N



2D illustration of the 4 potential solutions

Hartley & Zisserman 2003

Estimating F-matrix (the 8-point algorithm)



- Find multiple $\mathbf{X} \leftrightarrow \mathbf{X}'$ correspondences **(≥ 8)** between two images

$$\mathbf{x}'^{\mathsf{T}}\mathbf{F}\mathbf{x} = 0$$

$$x'xf_{11} + x'yf_{12} + x'f_{13} + y'xf_{21} + y'yf_{22} + y'f_{23} + xf_{31} + yf_{32} + f_{33} = 0$$

$$\mathbf{A}\mathbf{f} = \begin{bmatrix} x'_1x_1 & x'_1y_1 & x'_1 & y'_1x_1 & y'_1y_1 & y'_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \\ x'_nx_n & x'_ny_n & x'_n & y'_nx_n & y'_ny_n & y'_n & x_n & y_n & 1 \end{bmatrix} \mathbf{f} = \mathbf{0}$$

- **Enforce rank-2 constraint by SVD**

Hartley & Zisserman 2003

Normalized 8-point Algorithm

Objective

Given $n \geq 8$ image point correspondences $\{\mathbf{x}_i \leftrightarrow \mathbf{x}'_i\}$, determine the fundamental matrix $\mathtt{F}$ such that $\mathbf{x}'^{\mathsf{T}}_i \mathtt{F} \mathbf{x}_i = 0$.

Algorithm

(i) **Normalization:** Transform the image coordinates according to $\hat{\mathbf{x}}_i = \mathtt{T}\mathbf{x}_i$ and $\hat{\mathbf{x}}'_i = \mathtt{T}'\mathbf{x}'_i$, where $\mathtt{T}$ and $\mathtt{T}'$ are normalizing transformations consisting of a translation and scaling.

(ii) Find the fundamental matrix $\hat{\mathtt{F}}'$ corresponding to the matches $\hat{\mathbf{x}}_i \leftrightarrow \hat{\mathbf{x}}'_i$ by

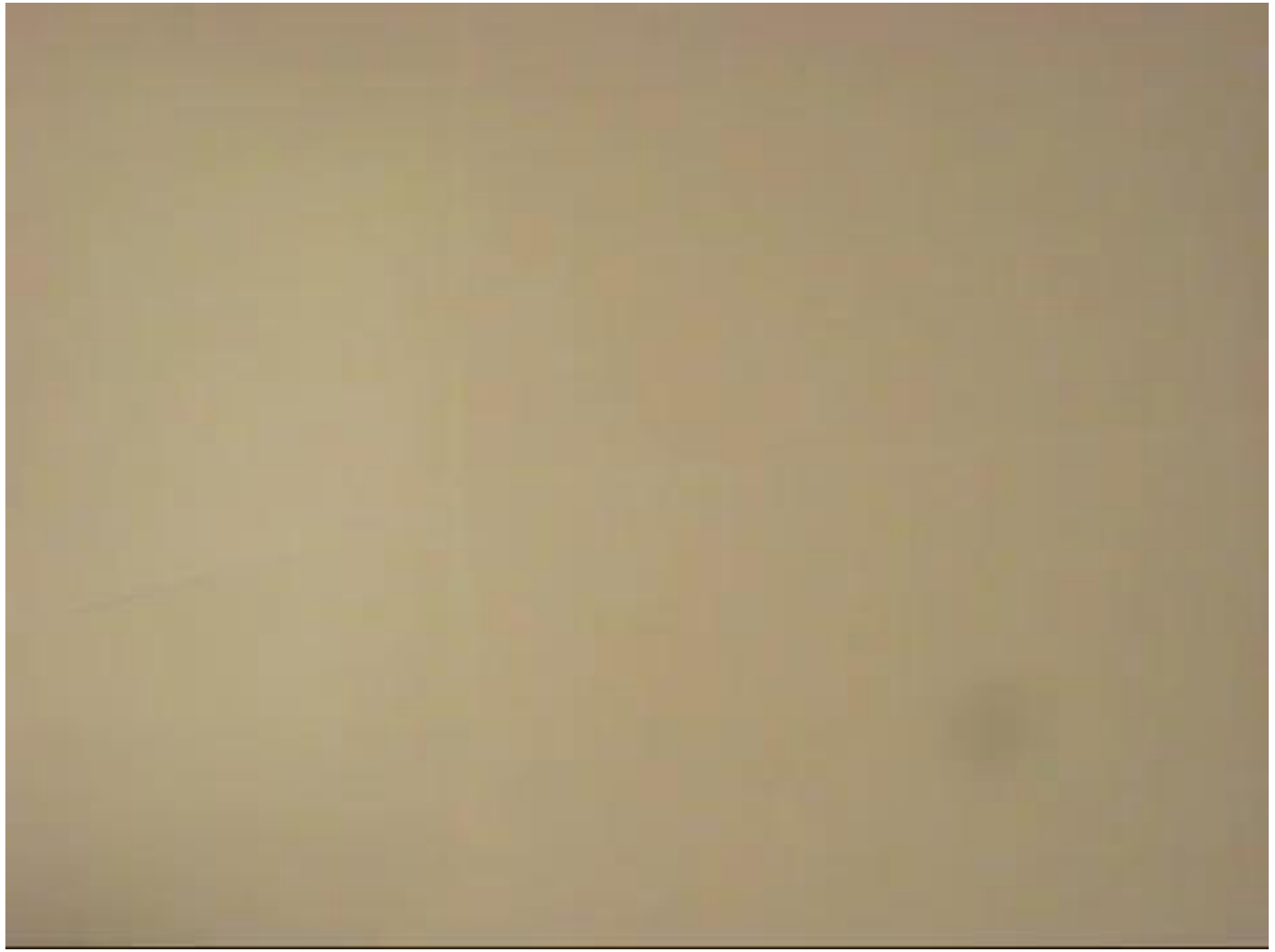
  (a) **Linear solution:** Determine $\hat{\mathtt{F}}$ from the singular vector corresponding to the smallest singular value of $\hat{\mathtt{A}}$, where $\hat{\mathtt{A}}$ is composed from the matches $\hat{\mathbf{x}}_i \leftrightarrow \hat{\mathbf{x}}'_i$ as defined in (11.3).

  (b) **Constraint enforcement:** Replace $\hat{\mathtt{F}}$ by $\hat{\mathtt{F}}'$ such that $\det \hat{\mathtt{F}}' = 0$ using the SVD (see section 11.1.1).

(iii) **Denormalization:** Set $\mathtt{F} = \mathtt{T}'^{\mathsf{T}} \hat{\mathtt{F}}' \mathtt{T}$. Matrix $\mathtt{F}$ is the fundamental matrix corresponding to the original data $\mathbf{x}_i \leftrightarrow \mathbf{x}'_i$.

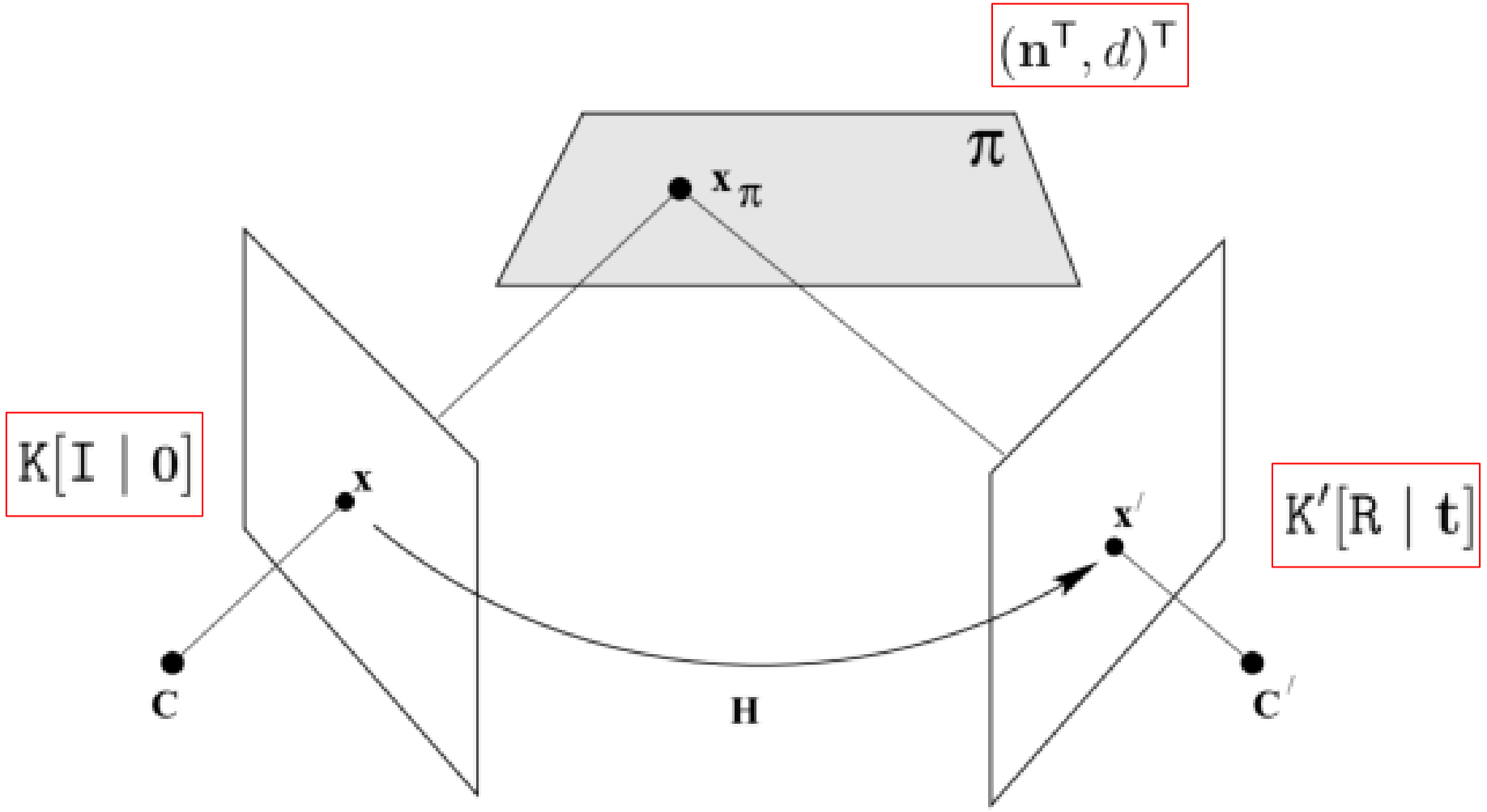
Hartley & Zisserman 2003

The Fundamental Matrix Song – Daniel Wedge



Video from: https://youtu.be/DgGV3l82NTk

Planar Homography



Hartley & Zisserman 2003

Planar Homography

- Any point X on the plane
  - Plane equation:

    $$n^T X + d = 0$$

  - Homogeneous coordinate:

    $$(X, \frac{-n^T X}{d})$$

- Image on camera 1:

  $$x_1 = KX$$

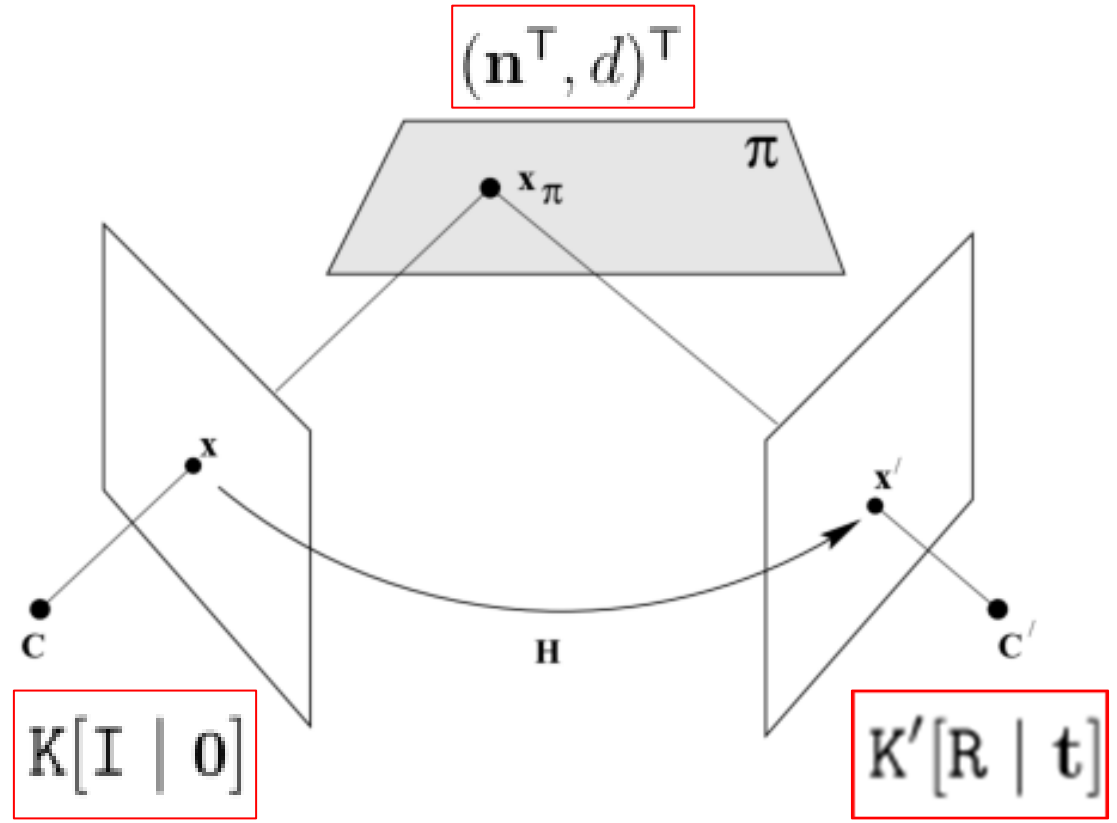
- Image on camera 2:

  $$x_2 = K'[RX - \frac{tn^T X}{d}]$$

- Relate two images by H

  $$x_2 = Hx_1 = HKX = K'[RX - \frac{tn^T X}{d}]$$

Hartley & Zisserman 2003



Malis, Ezio, and Manuel Vargas. "Deeper understanding of the homography decomposition for vision-based control." (2007).

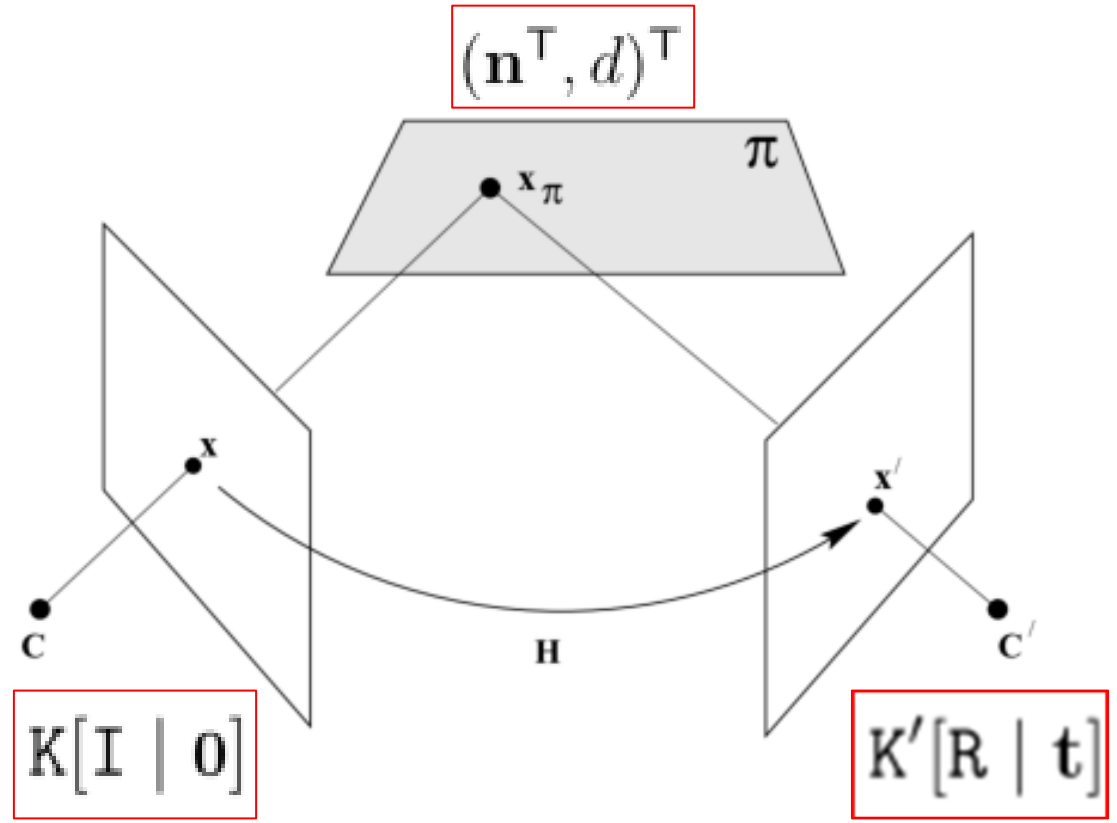
Planar Homography

- Above equation holds for any X

$$H = K'[RX - \frac{tn^T X}{d}]K^{-1}$$

$$K'^{-1}HK = R - \frac{tn^T}{d} = R - t'n^T$$

Decompose $\boldsymbol{K'^{-1}HK}$ to retrieve **R**, **t'**, and **n**



Malis, Ezio, and Manuel Vargas. "Deeper understanding of the homography decomposition for vision-based control." (2007).

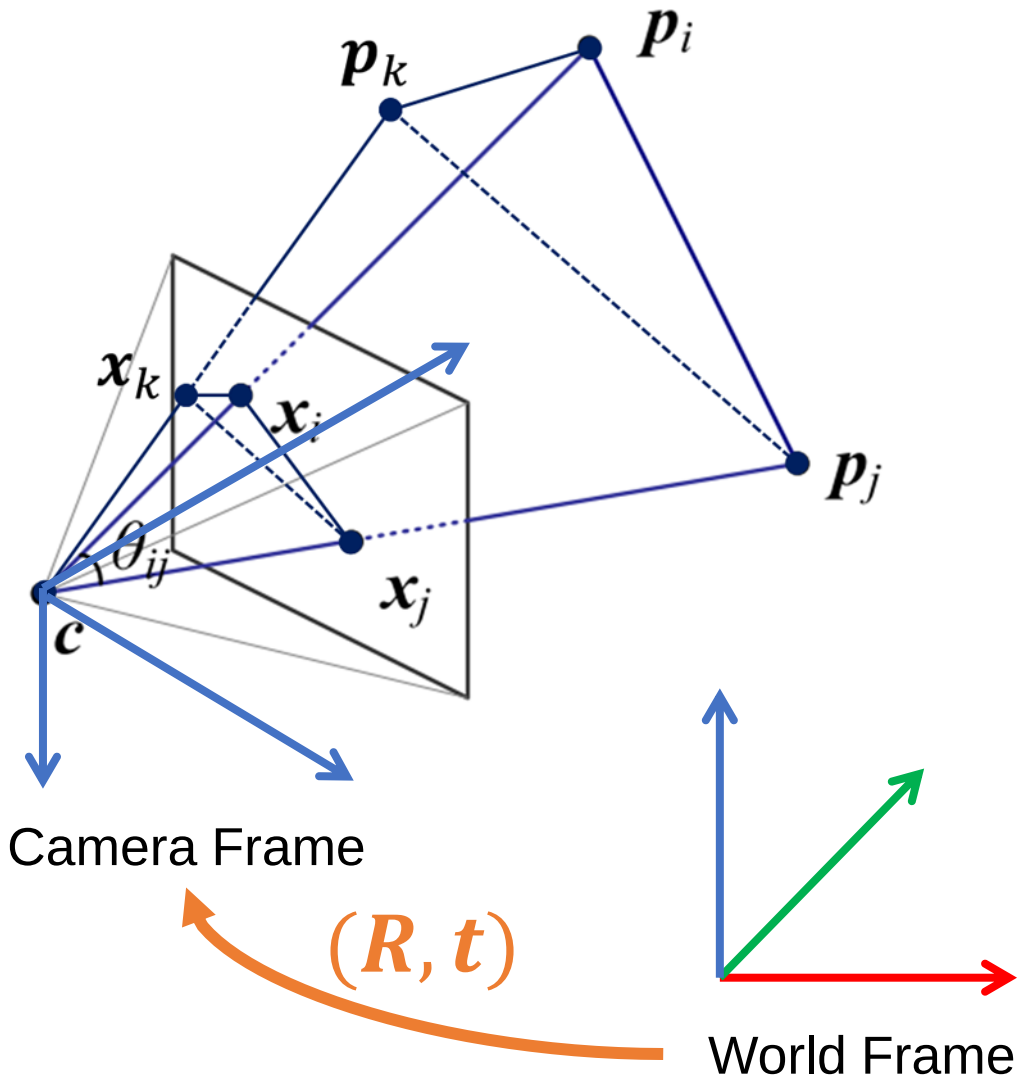
Hartley & Zisserman 2003

Perspective-n-point (PnP) Problem

- **Given/known variables**
  - $\boldsymbol{p}_i \in \mathbb{R}^3$, a set of n 3D points in the world frame
  - $\boldsymbol{x}_i \in \mathbb{P}^2$, a set of homogeneous image points
    - **corresponding to $\boldsymbol{p}_i$** (with the same index i) through camera projection
  - $\boldsymbol{K}$, camera intrinsics

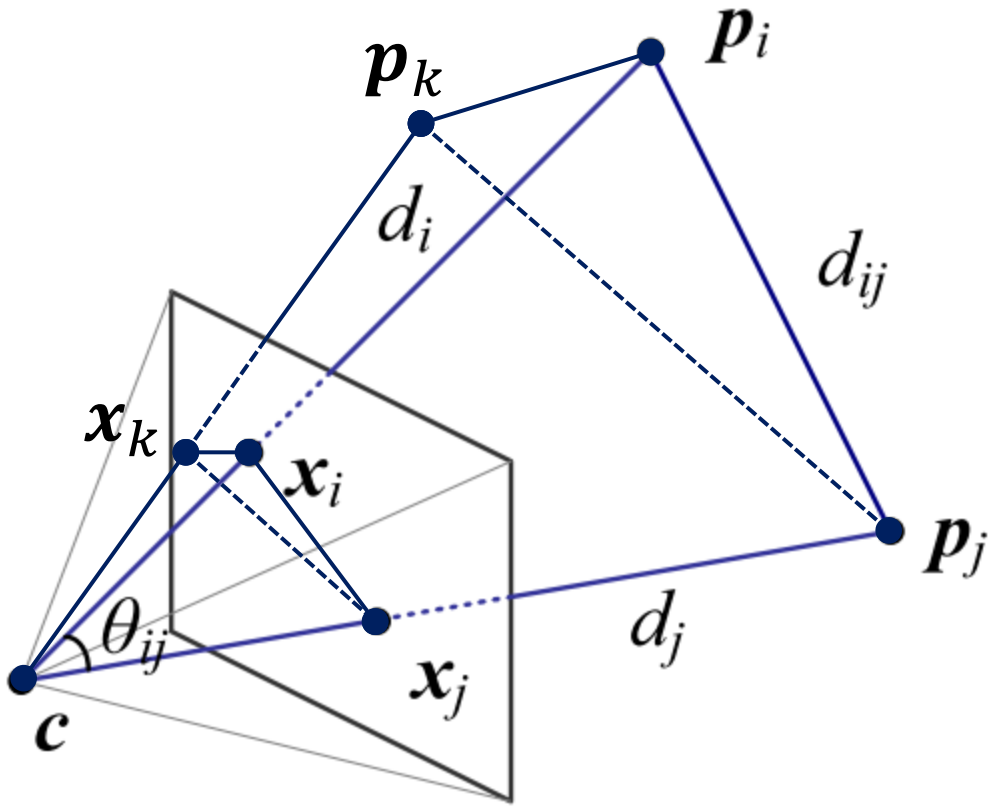
- **Unknown Variables**
  - $\boldsymbol{R}$, camera rotation
  - $\boldsymbol{t}$, camera translation
    - pose of the world frame w.r.t. the camera frame

- $s_i \boldsymbol{x}_i = \boldsymbol{K}(\boldsymbol{R}\boldsymbol{p}_i + \boldsymbol{t})$
  - Note: we can write "=" sign here instead of "$\propto$"
  - Because we write the unknown scale s_i explicitly



Perspective-n-point (PnP) Problem

- **A calibrated camera is an angular sensor**
  - $\hat{x}_i = K^{-1}x_i / \|K^{-1}x_i\|$
- Visual angles between any pair of image points must be the same as the angle between corresponding 3D points
  - $\angle x_i c x_j \equiv \angle p_i c p_j$
- 3 pair of 2D-3D correspondences leads to 4 possible solutions
  - P3P problem
- Useful for estimating object pose



Hand-eye Calibration

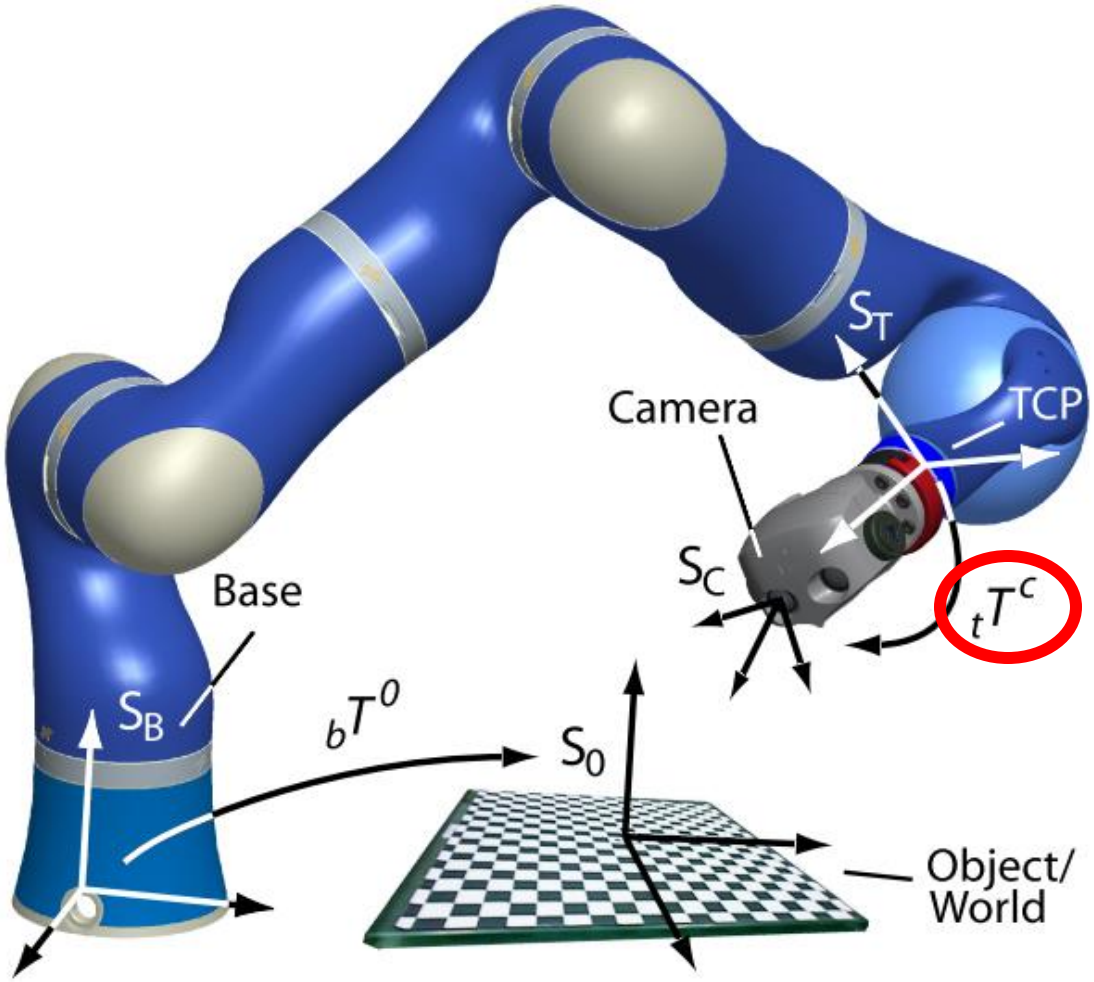


Image from Strobl & Hirzinger, ICRA'06

Hand-eye Calibration

- For any static point P in the calibration frame

$$B_1 X^{-1} A_1 P = B_2 X^{-1} A_2 P \quad ,\forall P$$

$$B_2^{-1} B_1 X^{-1} = X^{-1} A_2 A_1^{-1}$$

$$X B_1^{-1} B_2 = A_1 A_2^{-1} X$$

- $\mathbf{AX = XB}$

$$A = A_1 A_2^{-1}$$

$$B = B_1^{-1} B_2$$

- Solve X by observing multiple (A,B)

$$\begin{cases} A_{12}X & = & XB_{12} \\ & \vdots & \\ A_{i-1\,i}X & = & XB_{i-1\,i} \\ & \vdots & \\ A_{n-1\,n}X & = & XB_{n-1\,n} \end{cases}$$

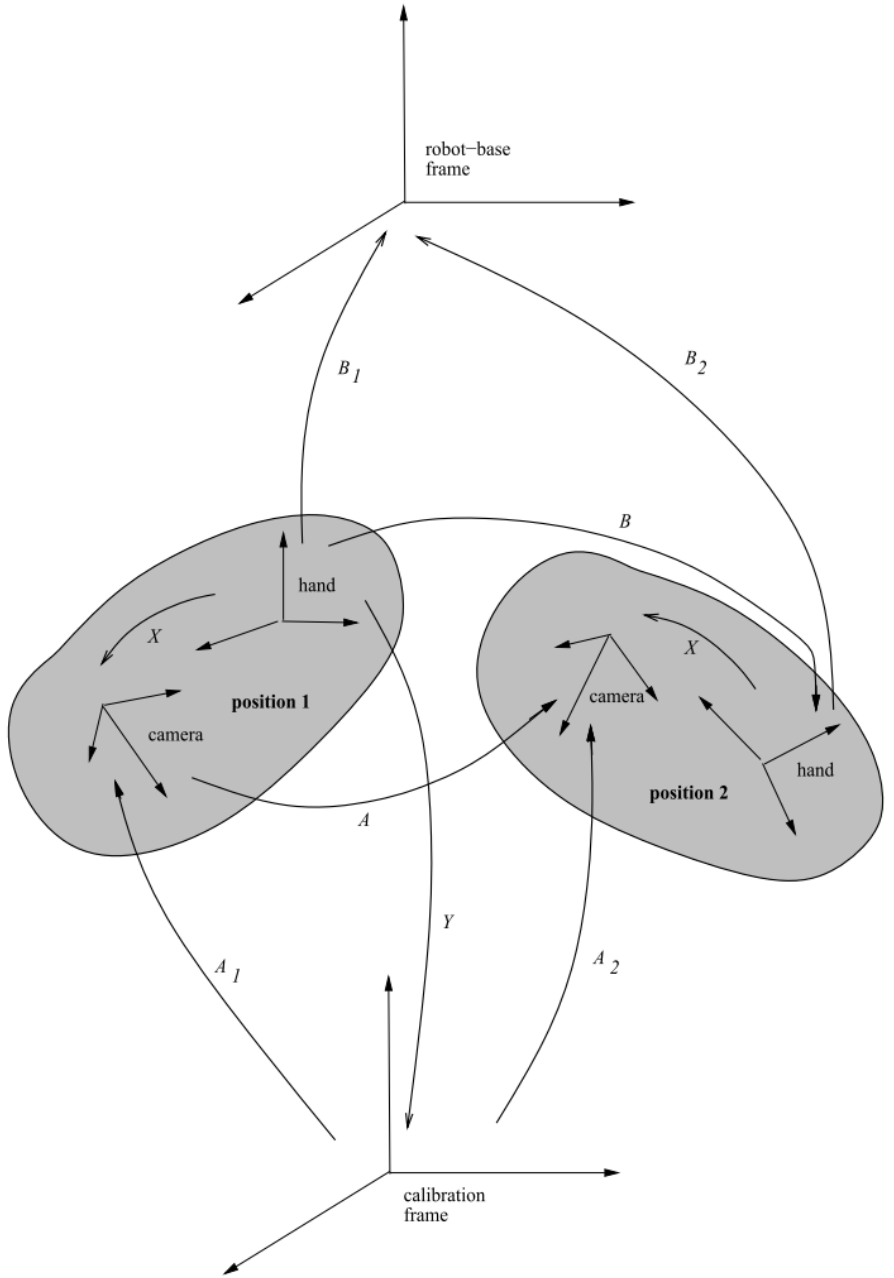


Image from Horaud 1995

Solve AX=XB

- $A = \begin{bmatrix} R_A & t_A \\ 0 & 1 \end{bmatrix}$, and so on for B and X
- $\begin{bmatrix} R_A & t_A \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_X & t_X \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_X & t_X \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_B & t_B \\ 0 & 1 \end{bmatrix}$

- $R_A R_X = R_X R_B$
- $R_A t_X + t_A = R_X t_B + t_X \Rightarrow (R_A - I) t_X = R_X t_B - t_A$

- Solve rotation first!

Solve $R_A R_X = R_X R_B$



- Let n_B be R_B's rotation axis
  - $n_B = R_B n_B$
- Multiply both sides by R_X
  - $R_X n_B = R_X R_B n_B$
  - $= R_A R_X n_B$
- Consider R_A's rotation axis: $n_A = R_A n_A$
- Since we have $R_X n_B = R_A R_X n_B$,
- … then $R_X n_B$ must be R_A's rotation axis
  - $\mathrm{n_A} = R_X n_B$
- R_X can be solved with multiple (≥ 2) pairs of (A, B)!
  - Polar decomposition on a covariance matrix $\boldsymbol{N}_A \boldsymbol{N}_B^T$
  - Orthogonal Procrustes problem

Orthogonal Procrustes Problem

$$R = \arg\min_{\Omega} \|\Omega A - B\|_F \quad \text{subject to} \quad \Omega^T\Omega = I,$$

$\boldsymbol{A}: 3 \times n, \boldsymbol{B}: 3 \times n, \boldsymbol{\Omega}: 3 \times 3$

$$\langle \mathbf{A}, \mathbf{B} \rangle_{\mathrm{F}} = \sum_{i,j} \overline{A_{ij}} B_{ij} \,, = \operatorname{tr}\left(\overline{\mathbf{A}^T}\mathbf{B}\right)$$

$$= \arg\min_{\Omega} \langle \Omega A - B, \Omega A - B \rangle$$

$$= \arg\min_{\Omega} \|\Omega A\|_F^2 + \|B\|_F^2 - 2\langle \Omega A, B \rangle$$

$$= \arg\min_{\Omega} \|A\|_F^2 + \|B\|_F^2 - 2\langle \Omega A, B \rangle$$

$$= \arg\max_{\Omega} \langle \Omega, BA^T \rangle$$

$$= \arg\max_{\Omega} \langle \Omega, U\Sigma V^T \rangle$$

$$= \arg\max_{\Omega} \langle U^T\Omega V, \Sigma \rangle$$

$$= \arg\max_{\Omega} \langle S, \Sigma \rangle \quad \text{where } S = U^T\Omega V$$

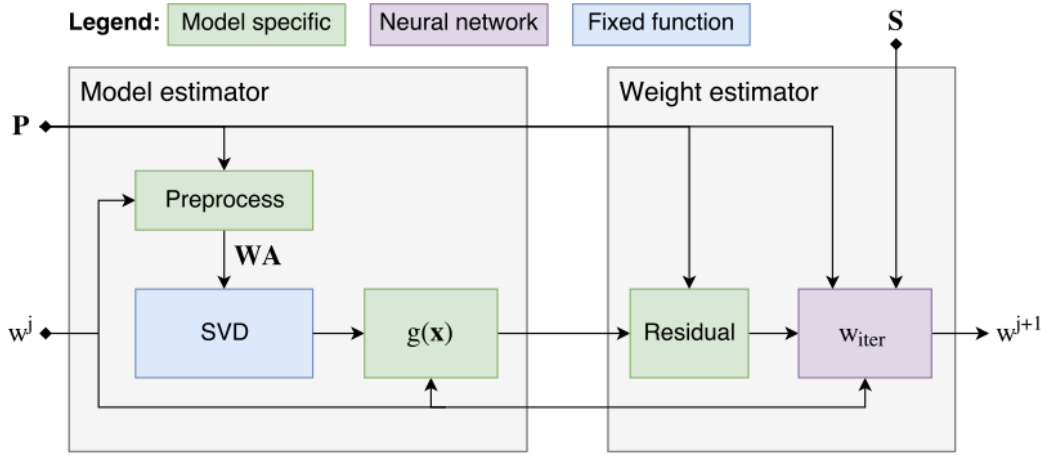
More generally, the trace is *invariant under cyclic permutations*, i.e.,

$$\operatorname{tr}(ABCD) = \operatorname{tr}(BCDA) = \operatorname{tr}(CDAB) = \operatorname{tr}(DABC).$$

$$S^* = I \implies I = U^T R V$$

$$R = UV^T$$

Recent Works on Multiview Geometry with Deep Learning



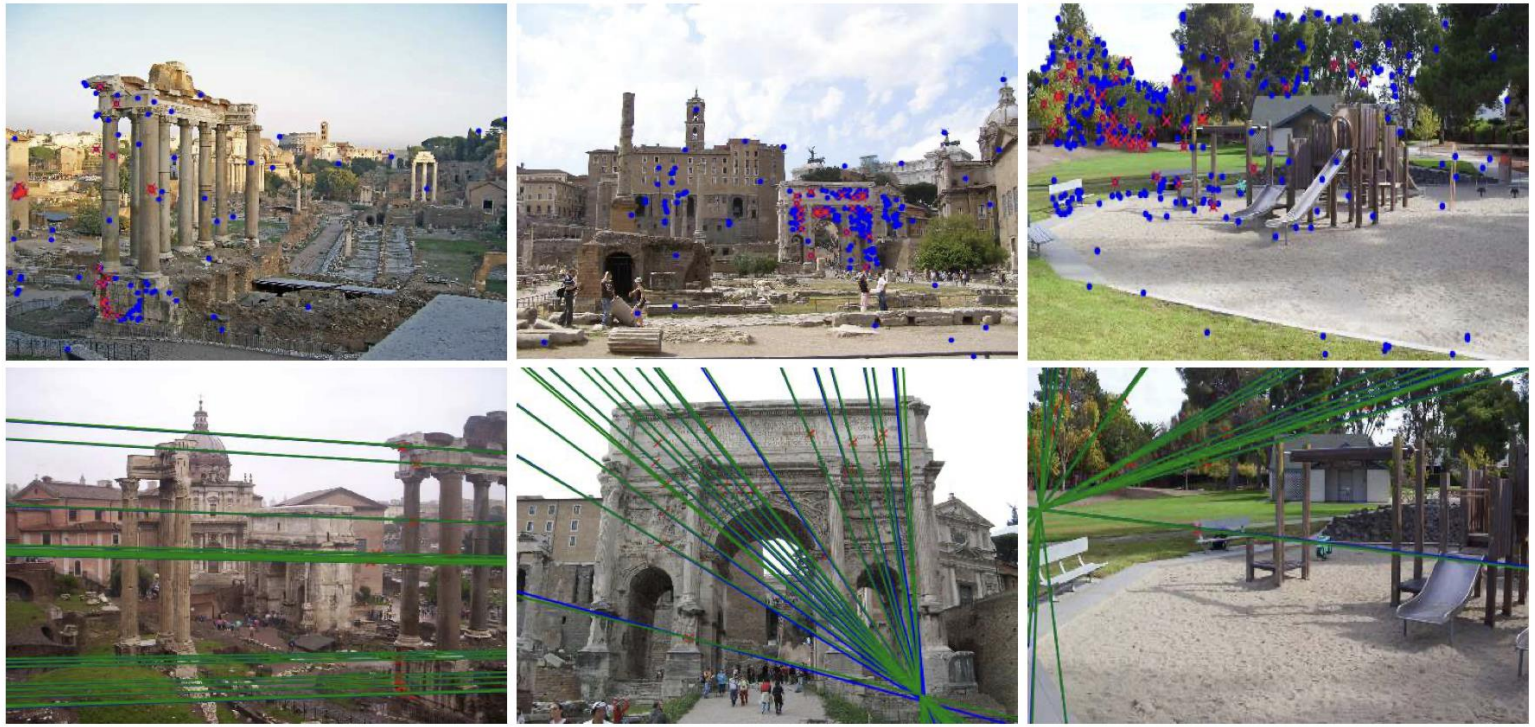
Layer	# in	# out	L-ReLU+IN
1	–	64	✓
2	64	128	✓
3	128	1024	✓
4	1024	512	✓
5	512	256	✓
6	256	1	✗

Figure 1 & Table 1. Estimation module and network architecture. Left: The estimation module is composed of two parts. Given input points and a weighting, a model is estimated using weighted least-squares. In the second stage a new set of weights is generated given the points, their residuals with respect to the previously estimated model, and possibly side information. Right: The network architecture of w_{init} and w_{iter}. A checkmark in column L-ReLU+IN indicates that a leaky ReLU followed by instance normalization is applied to the output of the layer.

The Challenge: Estimating Fundamental Matrix parameters via reweighted least-squares algorithm (IRLS) with a complex, learned reweighting function (parameterized by a NN basically)

Deep Fundamental Matrix Estimation, Rene et al., 2018

Recent Works on Multiview Geometry with Deep Learning



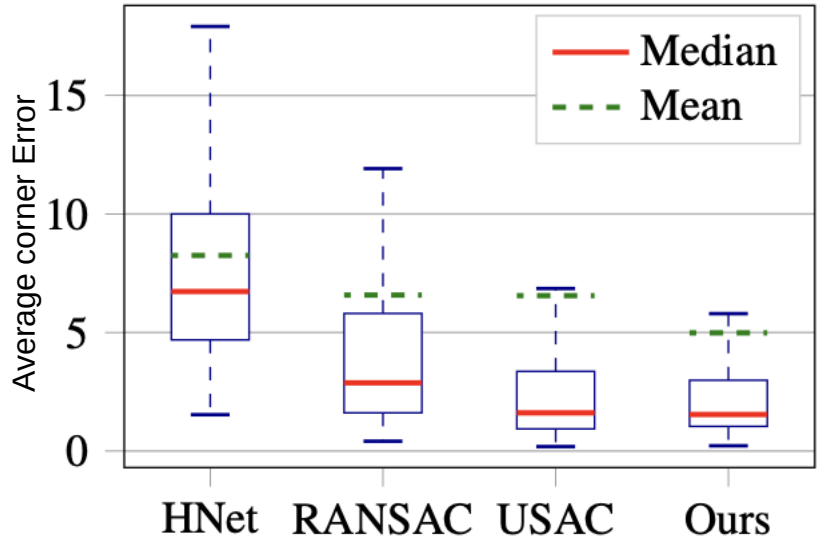


Figure 3. Image pairs from *Roman Forum* (first and second column) and *Tanks and Temples* (last column). Top row: First image with inliers (red) and outliers (blue). Bottom row: Epipolar lines of a random subset of inliers in the second image. We show the epipolar lines of our estimate (green) and of the groundtruth (blue). Images have been scaled for visualization.

Notes: It does quite well against traditional parameters estimation method. We can see some example of the Epipolar lines across corresponding views on the left.

Can we do better still? Yes!

Deep Fundamental Matrix Estimation, Rene et al., 2018

Recent Works on Multiview Geometry with Deep Learning

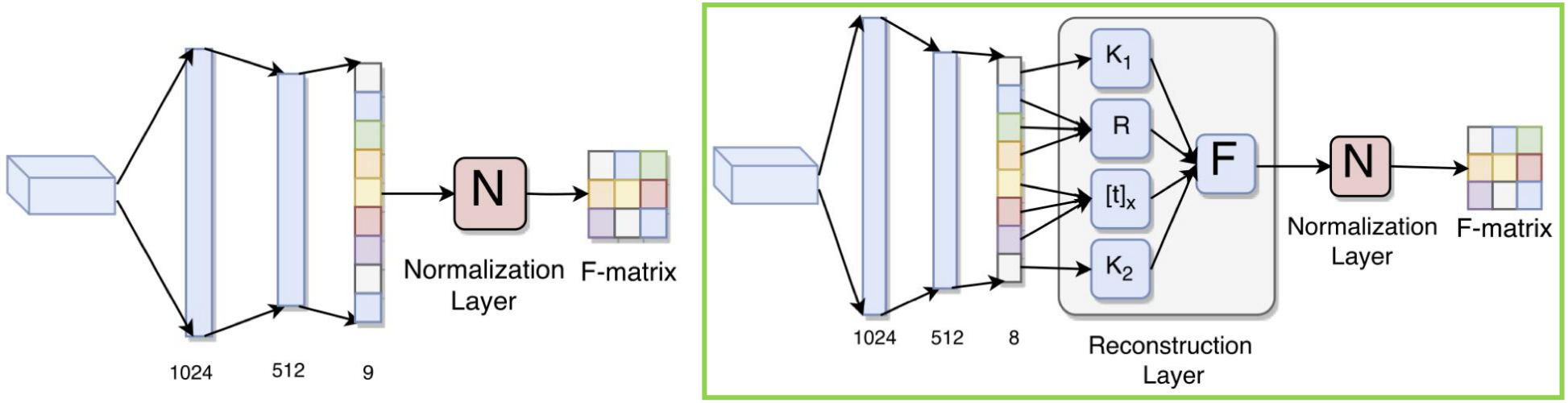


Fig. 3. Different regression methods for predicting F-matrix entries from the features. The architecture to directly regress the entries of the F-matrix is shown on the left. The network with the reconstruction and normalization layers is shown on the right, and is able to estimate homogeneous F-matrices with rank two and seven degrees of freedom.

The Idea:
1. Here we don't need correspondences to estimate the matrix, we just need the RGB input of Multiview images!
2. And instead of simple direct linear regression, they perform multiple regression via MLP layers specifically to reconstruct the individual components of the F-matrix to ensure proper structure – e.g. F matrix is of Rank 2.

Deep Fundamental Matrix Estimation without Correspondences, Omid et al., 2018

Recent Works on Multiview Geometry with Deep Learning

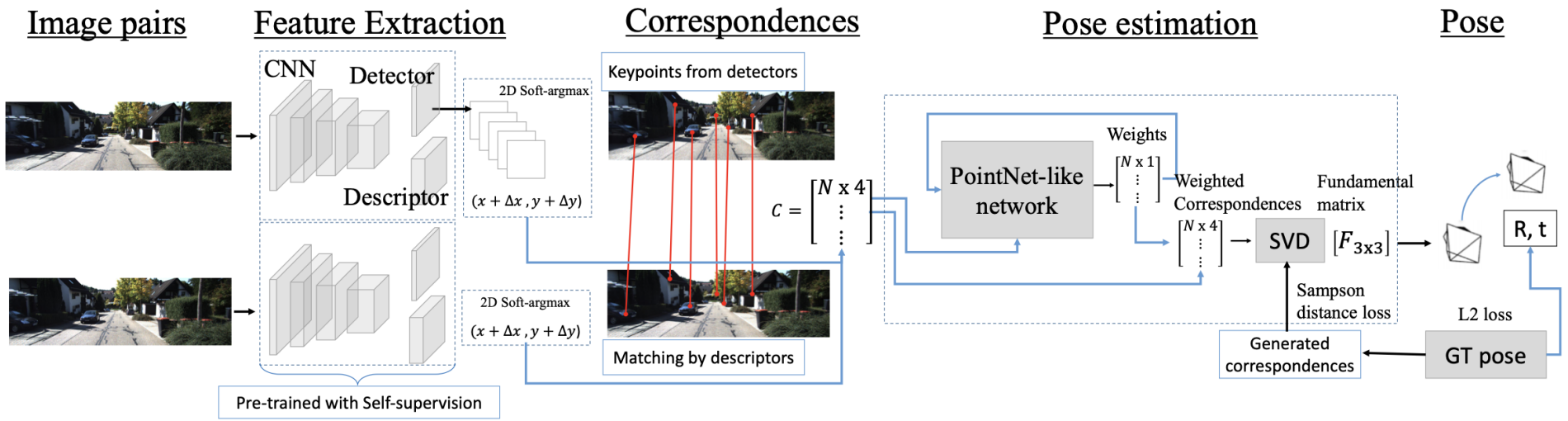


Fig. 1: **Overview of the system.** A pair of images is fed into the pipeline to predict the relative camera pose. Feature extraction predicts detection heatmaps and descriptors for finding sparse correspondences. Local 2D `Softargmax` is used as a bridge to get subpixel prediction with gradients. Matrix $\mathbf{C}$ of size $N \times 4$ is formed from correspondences. $\mathbf{C}$ is the input for pose estimation, where the PointNet-like network predicts weights for all correspondences. Weighted correspondences are passed through SVD to find fundamental matrix F, which is further decomposed into poses. Ground truth poses (GT poses) are used to compute L2 loss between rotation and translation (**pose-loss**). Correspondences generated from GT poses are used to compute fundamental matrix loss (**F-loss**). See more details in Sec. III.

The Idea:

1. Kind of a combination of prior methods – **Deep Feature Extraction** via SuperPoint + **Pose / F Matrix estimation** liken to ”Deep Fundamental Matrix Estimation” from slide 26.
2. End-to-end framework that performs feature extraction, correspondence matching and parameter estimation.

Deep Keypoint-Based Camera Pose Estimation with Geometric Constraints, You-Yi et al., 2020

Recent Works on Multiview Geometry with Deep Learning

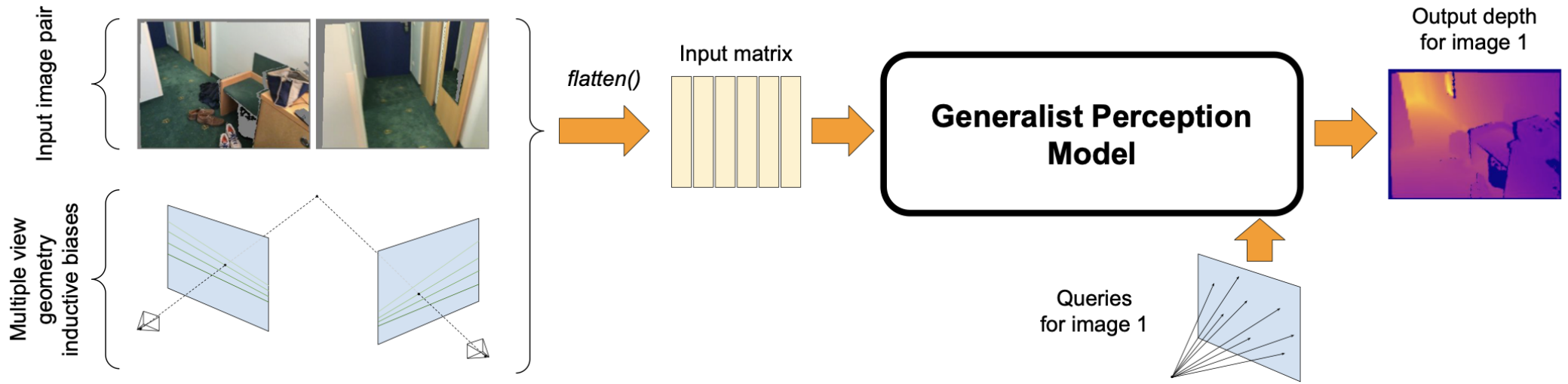


Figure 1. **Input-level inductive biases.** We explore 3D reconstruction using a generalist perception model, the recent Perceiver IO [23] which ingests a matrix of unordered and flattened inputs (*e.g.* pixels). The model is interrogated using a query matrix and generates an output for every query – in this paper the outputs are depth values for all pixels of the input image pair. We incorporate inductive biases useful for multiple view geometry into this generalist model without having to touch its architecture, by instead encoding them directly as additional inputs.

The Idea:
1. Here instead of estimating the intrinsic, the paper proposes a way to supply existing DL based vision models with handy information such as the aforementioned camera intrinsic as priors! (basically, the reverse problem from previous slides)
2. E.g. for Depth estimation with Perceiver IO (multipurpose vision model based with Transformers)

Input-level Inductive Biases for 3D Reconstruction, Wang et al., 2022

Overview

+ Feature detection
    - Harris/FAST/DoG
+ Feature description & matching
    - SIFT/SURF
++ Linear & total least square
* RANSAC
    - Intuitions behind RANSAC
    - How RANSAC works
    - Why minimal solution is important
    - More example problems
    - Variations

*: know how to code
++: know how to derive
+: know the concept

References

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